

Wumpus World Problem

4.1 Stench	4.2	4.3 Breeze	4.4 Pit
3.1 	Breeze Stench Gold	3.3 Pit	3.4 Breeze
2.1 Stench	2.2	2.3 Breeze	2.4
1.1 	1.2 Breeze	1.3 Pit	1.4 Breeze

Wumpus → Stench

Pit → Breeze

PEAS

① Performance Measure

- a) +1000 → agent comes out with gold
- b) -1 → every action
- c) -10 → using arrow
- d) -1000 → pit / wumpus
- e) Game ends: Agent = dead or climbs out of cave

② Environment

A 4x4 grid of rooms

↳ Agent = [1,1] Right facing

↳ locn. of Gold + Wumpus = random

↳ each square can start with pit $P=0.2$

③ Actuators

↳ motor: turn 90° left/right

↳ wheels: move forward

↳ Robotic arm: grab

↳ Robotic mech to shoot arrow

④ Sensors

a) Odor sensor — stench (adj of Wumpus)

b) Breeze perception

c) Camera — locate gold

d) Audio — Wumpus = killed → scream

e) Obstacle detection — Bump

Task env :

- ① Partially observable - state is not known fully - wumpus, pit, gold
- ② Deterministic - outcomes are known
- ③ Discrete - finite states, finite outcomes & moves
- ④ Sequential - order of moves matter
- ⑤ Static - world isn't evolving
- ⑥ Single Agent

Moves:

First Order Logic

Every
No

Some

$\forall x ($
 $\sim \exists x$

$\exists x ($

Q. All that glitters is not gold

$$\sim [\forall x \text{ glitter}(x) \rightarrow \text{gold}(x)]$$

$$\Rightarrow \sim [\forall x \sim \text{glitter}(x) \vee \text{gold}(x)]$$

$$\Rightarrow \exists x \text{ glitter}(x) \wedge \sim \text{gold}(x)$$

Q. All birds fly

$$\forall x : \text{birds}(x) \rightarrow \text{fly}(x)$$

Q. Some boys play cricket

$$\exists x : \text{boy}(x) \rightarrow \text{plays}(x, \text{cricket})$$

Q. Every student loves some student

$$\forall x (\text{student}(x) \rightarrow \exists y (\text{student}(y) \wedge \text{loves}(x, y)))$$

Q. Bill either takes analysis or geometry (but not both)

$A \Rightarrow$ Bill takes analysis $\text{takes}(\text{Bill}, \text{analy})$

$B \Rightarrow$ Bill takes geom. $\text{takes}(\text{Bill}, \text{geom})$

$$(A \vee B) \wedge \sim(A \wedge B)$$

Q. No student loves Bill

$\forall x: \text{student}(x) \rightarrow \sim \text{loves}(x, \text{Bill})$ or

$\sim \exists x: (\text{student}(x) \wedge \text{loves}(\text{Bill}))$

Q. Bill has atleast one sister

$$\exists x: \text{sister of}(x, \text{Bill})$$

Q. Bill has no sister

$\sim \exists x: \text{sister of}(x, \text{Bill})$

↑

there exist some

$\sim \Rightarrow$ no one

Q. Lucy is a prof.
 $\text{isprof}(\text{Lucy})$

Q. All prof are people
 $\forall x: (\text{isprof}(x) \rightarrow \text{people}(x))$

Q. John is the dean
 $\text{isdean}(\text{John})$

Q. Deans are prof.
 $\forall x: (\text{isdean}(x) \rightarrow \text{prof}(x))$

Q. All prof consider their dean as friend or don't know

$\forall x (\forall y (\text{isprof}(x) \wedge \text{isdean}(y) \rightarrow \text{friend}(x,y) \vee \sim \text{know}(x,y)))$

Q. Everyone is a friend of someone.
 $\forall x \exists y: \text{friend}(x,y)$

Q. People criticize only those who are not their friend

$\forall x (\forall y (\text{person}(x) \wedge \text{person}(y) \wedge \text{criticize}(x,y) \rightarrow \sim \text{friend}(y,x)))$

Q. is John no friend of Lucy?

is friend (John, Lucy)

Q. whoever can read is literate

$$\forall x: \text{can_read}(x) \rightarrow \text{is_literate}(x) \\ \sim \text{read}(x) \vee \text{literate}(x)$$

Q. Dolphins are not literate

$$\forall x: \text{dolphin}(x) \rightarrow \sim \text{literate}(x) \\ \sim \text{dolphin}(x) \vee \text{literate}(x)$$

Q. Some dolphins are intelligent

$$\exists x: [\text{dolphin}(x) \wedge \text{intelligent}(x)]$$

Q. Someone who is intelligent cannot read

$$\exists x: [\text{intelligent}(x) \wedge \sim \text{read}(x)]$$

$$\sim \Rightarrow \forall x (\sim \text{intelligent}(x) \vee \text{read}(x))$$

Q. Marcus was a man

→ man (Marcus)

Q. Marcus was a Pompeian

→ pompeian (Marcus)

Q. All pompeians are Romans

→ $\forall x : \text{pompeian}(x) \rightarrow \text{roman}(x)$

Q. Caesar was a ruler

→ ruler (Caesar)

Q. All Romans were either loyal to Caesar or hated him

$\forall x : \text{roman}(x) \rightarrow [\text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})]$

Q. Everyone is loyal to someone

$\forall x \exists y : \text{loyal}(x, y)$

Q. People only try to assassinate rulers they are not loyal to

$$\forall x \forall y : \text{people}(x) \wedge \text{ruler}(y) \wedge \text{assassinates}(x, y) \rightarrow \sim \text{loyal}(x, y)$$

Q. Marcus tried to assassinate Caesar
 $\text{try assass.}(\text{Marcus}, \text{Caesar})$

Q. All men are people

$$\forall x : \text{man}(x) \rightarrow \text{person}(x)$$

Q. Gorilla is Black

$$\text{gorilla}(x) \rightarrow \text{black}(x)$$

Q. All Kings are persons

$$\forall x: \text{king}(x) \rightarrow \text{person}(x)$$

Q. Every city in Maharashtra has a temple

$$\forall x: \text{city}(x) \wedge \text{inMaha}(x) \rightarrow \text{has temple}(x)$$

Q. An apple a day keeps doctor away

$$\forall x \forall y ((\text{Person}(x) \wedge \text{Apple}(y) \wedge \text{eats}(x, y)) \rightarrow \neg \text{need doctor}(x))$$

Q. Anything anyone eats and is not killed by is food

$$\forall x \forall y: \text{person}(x) \wedge \text{eats}(x, y) \wedge \neg \text{killed}(x)$$

→ food (y)

Q. Square of 3 is 9

square of (9, 3)

Resolution —

- ① convert facts $\rightarrow$ FOL
- ② FOL to CNF
- ③ Negate result
- ④ Draw resolution graph

Q.

a. John likes all kind of food

$$\forall x : \text{food}(x) \rightarrow \text{likes}(\text{John}, x)$$

b. Apple and vegetable are food

$$\text{food}(\text{Apple}) \wedge \text{food}(\text{vegetable})$$

c. Anything anyone eats and is not killed by is food

$$\forall x \forall y : \text{person}(\cancel{x}) \text{ eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$$

d- Anil eats peanuts and still alive

$\text{eats}(\text{Anil}, \text{peanuts}) \wedge \text{alive}(\text{Anil})$

e- Harry eats everything that Anil eats

$\forall x : \text{eats}(\text{Anil}, x) \rightarrow \text{eats}(\text{Harry}, x)$

Steve only likes easy courses

$\hookrightarrow \forall x: \text{easy}(x) \rightarrow \text{likes}(\text{Steve}, x)$

x

~~①~~ $\hookrightarrow \neg \text{easy}(x) \vee \text{likes}(\text{Steve}, x)$

Science courses are hard

$\hookrightarrow \forall x: \text{science}(x) \rightarrow \neg \text{easy}(x)$

② $\hookrightarrow \neg \text{science}(x) \vee \neg \text{easy}(x)$

All courses in the basket-weaving dept are easy

$\hookrightarrow \forall x: \text{inbasket-weaving}(x) \rightarrow \text{easy}(x)$

~~③~~ $\hookrightarrow \neg \text{inbw}(x) \vee \text{easy}(x)$

y

BK301 is a basket-weaving course

$\hookrightarrow \text{inbasket-weaving}(\text{BK301})$

④ $\text{inbw}(\text{BK301})$ x

$\Rightarrow$ what course does Steve like?

$\text{likes}(\text{Steve}, x)$

$\hookrightarrow \neg \text{likes}(\text{Steve}, x)$

$\sim \text{likes}(\text{Steve}, x)$

$\neg \text{easy}(x) \vee \text{likes}(\text{Steve}, x)$

$\neg \text{easy}(x)$

$\sim \text{inbw}(x) \vee \neg \text{easy}(x)$

$\neg \text{inbw}(x)$

$\text{inbw}(\text{BK301})$

$x/\text{BK301}$

null

$\therefore x = \underline{\underline{\text{BK301}}}$

①

Q. Marcus was a man

→ man (Marcus)

~~②~~

Q. Marcus was a Pomp

→ pompeian (Marcus)

~~③~~

Q. All pompeians are Romans

→ $\forall x: \text{pompeian}(x) \rightarrow \text{roman}(x)$

$\sim \text{pomp}(x) \vee \text{roman}(x)$

~~④~~

Q. Caesar was a ruler

→ ruler (Caesar)

~~⑤~~

Q. All Romans were either loyal to Caesar or hated him

$\forall x: \text{roman}(x) \rightarrow [\text{loyal}(x, \text{Caesar}) \vee \text{hate}(x, \text{Caesar})]$

$\sim \text{roman}(x) \vee \text{loyal}(x, c) \vee \text{hate}(x, c)$

⑥

Everyone is loyal to someone

$\forall x (\exists y): \text{loyal}(x, y)$

loyal(x, f(x))

~~7~~ Q. People only try to assassinate ruler
are not loyal to

$\forall x \forall y : \text{people}(x) \wedge \text{ruler}(y) \wedge$
 $\text{assassinate}(x, y) \rightarrow$
 $\neg \text{loyal}(x)$

$\neg \text{people}(x) \vee \neg \text{ruler}(y) \vee$
 $\neg \text{ass.}(x, y) \vee \neg \text{loyal}(x)$

~~8~~ Marcus tried to assassinate Caesar
 $\text{tryassn.}(\text{Marcus}, \text{Caesar})$

~~9~~ Q. All men are people
 $\forall x : \text{man}(x) \rightarrow \text{person}(x)$
 $\neg \text{man}(x) \vee \text{person}(x)$

$\Rightarrow \neg \text{hate}(\text{Marcus}, \text{Caesar})$

~~$\neg \text{hate}(M, c)$~~ $\neg \text{roman}(x)$ $\neg \text{loyal}(x, c)$ ~~$\neg \text{hate}(x, c)$~~

marcus / x

~~$\neg \text{roman}(m)$~~ $\vee \text{loyal}(m, c)$

~~$\neg \text{pom}(x)$~~ $\vee$ ~~$\neg \text{roman}(x)$~~

marcus / x

$\text{loyal}(m, c)$ $\vee$ ~~$\neg \text{pom}(m)$~~

~~$\text{pom}(x)$~~

m / x

~~$\text{loyal}(m, c)$~~

$\neg \text{people}(x) \vee \neg \text{ruler}(y)$

$\vee \neg \text{anc}(x, y) \vee$

~~$\neg \text{loyal}(x)$~~

$\neg \text{people}(m) \vee \neg \text{ruler}(c) \vee$ ~~$\neg \text{anc}(m, c)$~~

~~$\neg \text{anc}(m, c)$~~

$\neg \text{people}(m) \vee$ ~~$\neg \text{ruler}(c)$~~

~~$\text{ruler}(c)$~~

~~$\neg \text{people}(m)$~~
person

~~$\neg \text{man}(x) \vee \text{person}(x)$~~

$\neg \text{man}(x)$

$\text{man}(m)$

null

$\therefore \text{false}$

① All people who are earning are happy

$$\forall x : \text{earning}(x) \rightarrow \text{happy}(x)$$

$$\Rightarrow \neg \text{earning}(x) \vee \text{happy}(x)$$

② All happy people smile

$$\forall x : \text{happy}(x) \rightarrow \text{smile}(x)$$

$$\Rightarrow \neg \text{happy}(x) \vee \text{smile}(x)$$

③ Someone is earning

$$\exists x : \text{earning}(x)$$

$$\Rightarrow \text{earning}(E(a))$$

$$\neg \exists x : \text{smiling}(x)$$

$$\forall x : \neg \text{smiling}(x) = \neg \text{smiling}(x)$$

$$\neg \text{smiling}(x)$$

$$\neg \text{happy}(x) \vee \text{smiling}(x)$$

$$\neg \text{happy}(x)$$

$$\neg \text{earning}(x) \vee \text{happy}(x)$$

$$\neg \text{earning}(x)$$

$$\text{earning}(E(a))$$

$$E(a) / x$$

$\therefore$ proved.

rule

Forward Chaining

↳ based on data availability → decision taken

Q. 1) It is crime for an American to sell weapons to the enemy of America

American(x) ∧ weapon(y) ∧ enemy(z, America) ∧
sells(x, y, z) → Criminal(x) ~~x~~

2) Country nano is an enemy of America

enemy(Nano, America)

~~sells x~~
~~fact~~

3) Nano has some missiles

∃x : missiles(x) ∧ owns(nano, x)

4) All of its missiles were sold to it by Col. West

∀x : missiles(x) ∧ owns(Nano, x) →

sold (Col. West, x ,
Nano)

5) Col. West who is an American

American (Col. West)

6 missiles(x) $\rightarrow$ weapon(x)

$\rightarrow$ lit fact (LHS)

American (Col. West) missile(x) born(Nano, x)
enemy(Nano, America)

\ Root $\rightarrow$ check if any statement's LHS present

Q-

1. Fred is a collie
collie (Fred)

2. Sam is Fred's master
master (Fred, Sam)

3. The day is Saturday
day (Saturday)

4. It is cold on Sat
cold (Sat.)

5. Fred is trained
trained (Fred)

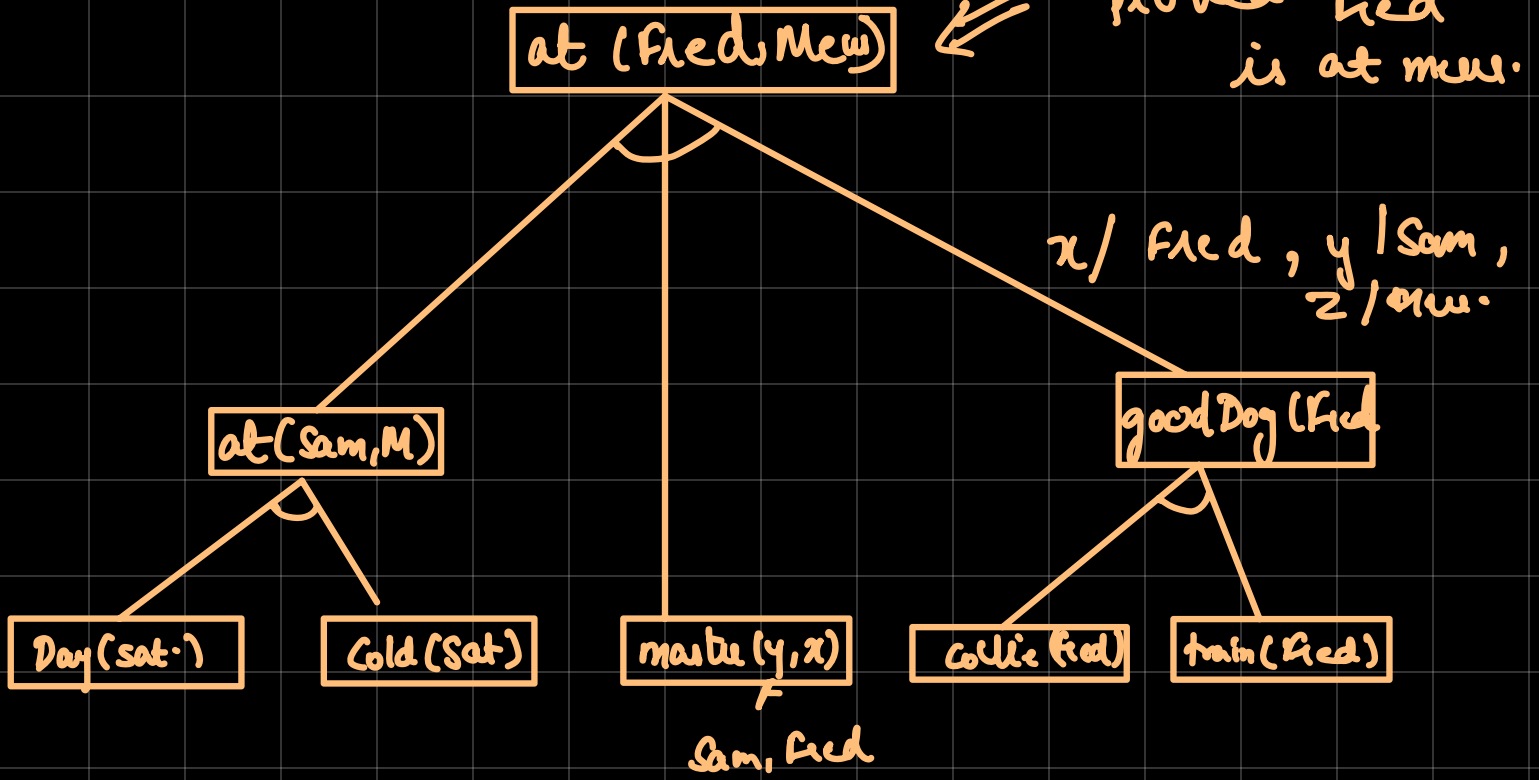
6. Trained collies are good dogs
 $\forall x: \text{collie}(x) \wedge \text{trained}(x) \rightarrow \text{good dog}(x)$

7. If a dog is good & has a master and at some place, then he will be with his master

$\forall x \forall y \forall z: \text{goodDog}(x) \wedge \text{master}(y, x) \wedge \text{at}(y, z) \rightarrow \text{at}(x, z)$

8. If day is sat. & day is cold, then Sam is at museum

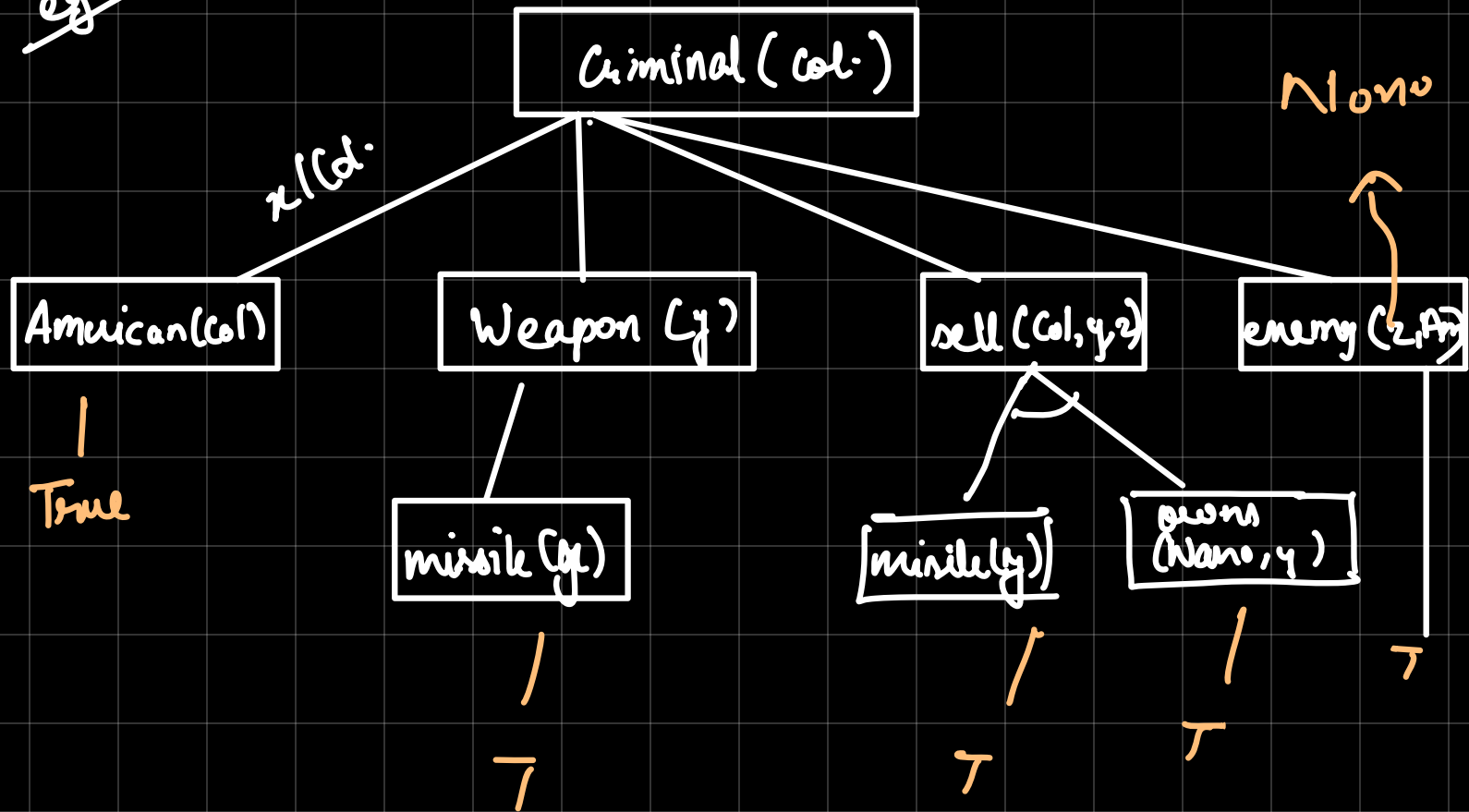
$\text{day}(\text{sat.}) \wedge \text{cold}(\text{Sat}) \rightarrow \text{at}(\text{sam}, \text{museum})$



Backward Chaining

Goal $\rightarrow$ Fact

America
eg



- Q.
1. Every student smiles
 2. No one talks
 3. At least one student failed History
 4. Every person who buys an insurance policy is smart
 5. No person buys an expensive policy

$$1) \quad \forall x: \text{student}(x) \rightarrow \text{smiles}(x)$$

$$2) \quad \exists x: \text{talks}(x) \Rightarrow \sim \exists x \text{ talks}(x) \\ = \forall x \sim \text{talks}(x)$$

$$3) \quad \exists x: (\text{student}(x) \wedge \text{fail}(x, \text{History}))$$

$$4) \quad \forall x: \text{person}(x) \wedge \text{buys}(x, \text{insurance}) \rightarrow \text{smart}(x)$$

$$5) \quad \exists x: \text{buys}(x, \text{exp. policy}) \Rightarrow \sim [\quad]$$

$$\forall x: \sim \text{buys}(x, \text{exp. policy})$$

Q. a) Ravi likes all kind of food

b) Apple & Chicken are food

c) Anything anyone eats and is not killed is food.

d) Ajay eats peanuts & still alive.

e) Rita everything that Ajay eats.

b) ~~a)~~ $\forall x: \text{food}(x) \rightarrow \text{likes}(\text{Ravi}, x)$
 $\neg \text{food}(x) \vee \text{likes}(\text{Ravi}, x)$

b) $\text{food}(\text{Apple}) \wedge \text{food}(\text{Chicken})$
 $\hookrightarrow ,$

~~c)~~ $\forall x \forall y: \text{eats}(x, y) \wedge \neg \text{killed}(x) \rightarrow \text{food}(y)$
 $\neg \text{eats}(x, y) \vee \text{killed}(x) \vee \text{food}(y)$

~~d)~~ $\text{eats}(\text{Ajay}, \text{peanuts}) \wedge \text{alive}(\text{Ajay})$

e) $\forall x: \text{eats}(\text{Ajay}, x) \rightarrow \text{eats}(\text{Rita}, x)$
 $\neg \text{eats}(\text{Ajay}, x) \vee \text{eats}(\text{Rita}, x)$

$alive(x) \rightarrow \neg killed(x)$ $\neg killed(x) \rightarrow alive(x)$
 $\neg alive(x) \vee \neg killed(x)$ $killed(x) \vee alive(x)$

GOAL = Ravi likes Peanuts

$\hookrightarrow likes(Ravi, Peanuts)$

Negat $\neg likes(Ravi, Peanuts)$

